

# The Koide $C_3$ Closure — Tautology, Saddle Point, and the Real Problem

**The spacing is automatic. The potential is a saddle. The amplitude is the question.**

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**Backed by:** DATA-3 (32/32 checks), PHYS-8 (Koide parametrization), inline computation

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## Abstract

The Koide formula  $K = (m_e + m_\mu + m_\tau) / (\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$  has held for charged leptons to six significant figures since 1982. The natural attempt to derive it is through  $C_3$  symmetry: the three phases in the Koide parametrization  $\sqrt{m_k} = M(1 + a \cos(\theta_0 + 2\pi k/3))$  are equally spaced at  $120^\circ$ , which looks like the ground state of a frustrated  $C_3$  potential. This paper proves the  $C_3$  path is dead, with two independent arguments either of which is sufficient. First, the  $120^\circ$  spacing is a tautology — three parameters ( $M, a, \theta_0$ ) fitting three data points ( $m_e, m_\mu, m_\tau$ ) is an exactly determined system, and ANY three positive masses have this form. The spacing is a property of the parametrization, not of the physics. Second,  $K = 2/3$  is a saddle point of the Koide ratio evaluated on the  $C_3$  phase landscape — perturbing the phases in one direction increases  $K$ , in another direction decreases it. The  $C_3$  potential does not select  $K = 2/3$  as a preferred value. The path is doubly dead: its success ( $120^\circ$  spacing) is tautological, and its failure (saddle point, no amplitude selection) is on the only thing that matters — why  $a^2 = 2$  for charged leptons. The real problem is the amplitude. Every known reformulation of  $K$

= 2/3 (as  $a = \sqrt{2}$ ,  $CV(\sqrt{m}) = 1$ ,  $Var(\sqrt{m}) = Mean(\sqrt{m})^2$ , midpoint of allowed range) is algebraically equivalent. None is a derivation. The open question is: derive  $a^2 = 2$  from physics, or explain why charged leptons satisfy it while quarks do not ( $a^2_{\text{down}} = 2.39$ ,  $a^2_{\text{up}} = 3.09$ ).

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## 1. The Koide Formula

The Koide formula is about the rest masses of the three charged leptons — the electron, muon, and tau. It is unrelated to the anomalous magnetic moment computed in PHYS-9 and PHYS-22, despite involving the same particles. It concerns masses, not magnetic properties.

Yoshio Koide proposed in 1982 that:

$$K = (m_e + m_\mu + m_\tau) / (\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$$

Using DATA-3 masses ( $m_e = 0.51100$  MeV,  $m_\mu = 105.658$  MeV,  $m_\tau = 1776.86$  MeV), the left side evaluates to 0.666661, deviating from  $2/3 = 0.666667$  by  $6 \times 10^{-6}$ . The deviation is  $0.91\sigma$  given the measurement uncertainty on the tau mass. The relation is consistent with being exact at the available precision, which is limited by the tau mass at five significant figures.

No derivation from known physics exists. The formula has held for over 40 years through multiple improvements in the tau mass measurement. At every stage, the ratio has remained within measurement uncertainty of  $2/3$ . It is the most precise empirical mass relation in particle physics.

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## 2. The Parametrization

### The Koide Circle: Charged Leptons in $\sqrt{m}$ Space

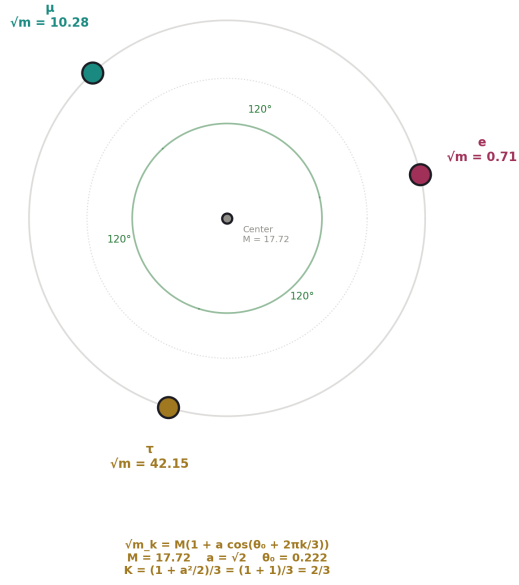


Figure 1: Fig. 3:  $\sqrt{m}_e, \sqrt{m}_\mu, \sqrt{m}_\tau$  placed on a circle at  $120^\circ$  intervals in  $\sqrt{m}$  space.  $M = 17.72$ ,  $a = \sqrt{2}$ ,  $\theta_0 = 0.222$ . The parametrization  $K = (1+a^2/2)/3 = 2/3$  when  $a^2 = 2$ .

The three charged lepton masses can be written in the Koide parametrization:

$$\sqrt{m}_k = M(1 + a \cos(\theta_0 + 2\pi k/3)), \text{ for } k = 0, 1, 2$$

Three parameters ( $M, a, \theta_0$ ) for three masses ( $m_e, m_\mu, m_\tau$ ). The scale  $M = (\sqrt{m}_e + \sqrt{m}_\mu + \sqrt{m}_\tau)/3$  is the mean of the square roots. The amplitude  $a$  measures the spread around the mean. The phase  $\theta_0$  sets the orientation.

PHYS-8 established the general identity: for  $N$  equally-spaced objects on a circle in  $\sqrt{m}$  space, the ratio  $(\Sigma m)/(\Sigma \sqrt{m})^2 = (1 + a^2/2)/N$ . This is a trigonometric identity valid for  $N \geq 3$ , proved from the facts that  $N$  equally-spaced cosines sum to zero and the sum of their

squares equals  $N/2$ .

For charged leptons ( $N = 3$ ):  $K = (1 + a^2/2)/3$ . So  $K = 2/3$  is equivalent to  $a^2 = 2$ . These are the same statement in different variables.

The three-sector data from DATA-3:

| Sector                               | K       | $a^2$ | $K - 2/3$             |
|--------------------------------------|---------|-------|-----------------------|
| Charged leptons (e, $\mu$ , $\tau$ ) | 0.66666 | 2.000 | $-6 \times 10^{-6}$   |
| Down quarks (d, s, b)                | 0.7313  | 2.388 | $+6.5 \times 10^{-2}$ |
| Up quarks (u, c, t)                  | 0.8488  | 3.093 | $+1.8 \times 10^{-1}$ |

Only charged leptons satisfy  $K = 2/3$ . The quark sectors deviate by 10% and 27% respectively. The amplitude ordering is  $a^2_{\text{lep}} < a^2_{\text{down}} < a^2_{\text{up}}$ .

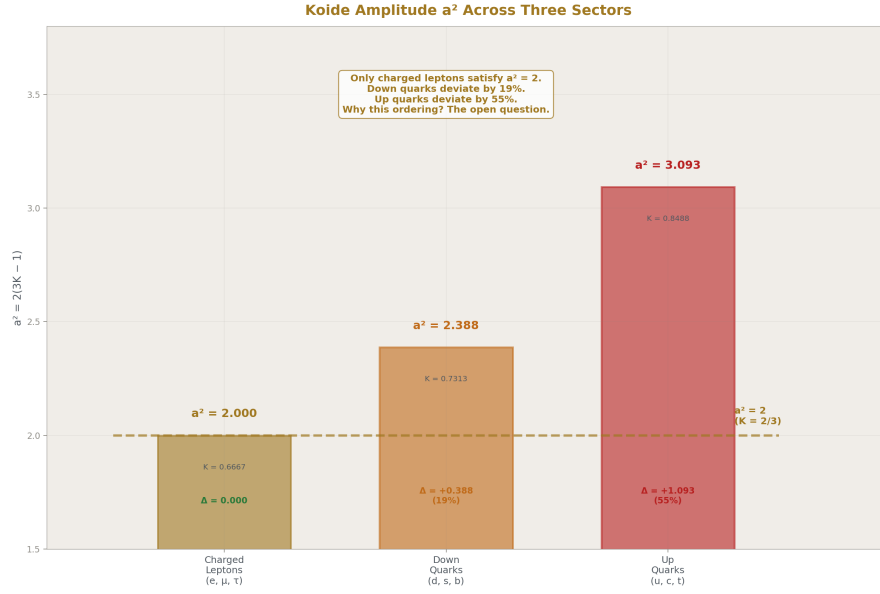


Figure 2: Fig. 5:  $a^2 = 2.000$  for charged leptons (exact), 2.388 for down quarks (19% off), 3.093 for up quarks (55% off). Only leptons satisfy  $K = 2/3$ . The ordering must be explained by any future theory.

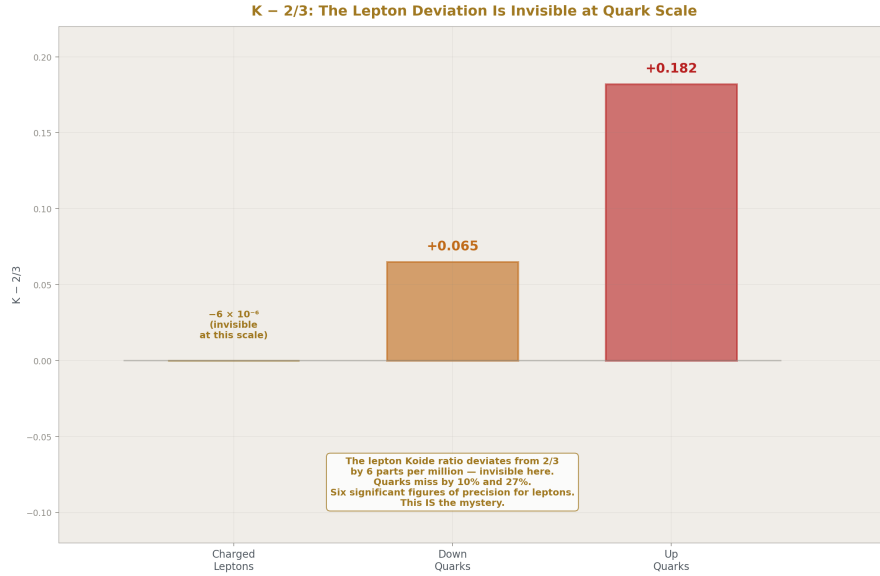


Figure 3: Fig. 6: Lepton K deviates from 2/3 by 6 parts per million — invisible at the scale of quark deviations (10% and 27%). Six significant figures of precision for leptons IS the mystery.

### 3. The Tautology

The  $120^\circ$  spacing in the Koide parametrization is not a discovery. It is not a prediction. It is not physics. It is a tautology.

The proof is one sentence: three parameters fitting three data points is an exactly determined system.

The parametrization  $\sqrt{m_k} = M(1 + a \cos(\theta_0 + 2\pi k/3))$  has three unknowns ( $M$ ,  $a$ ,  $\theta_0$ ) and three equations (one for each mass). Three equations in three unknowns generically have a unique solution. This means ANY three positive masses can be written in this form. The  $120^\circ$  spacing is built into the parametrization by construction — it is the coordinate choice, not a property of the data.

The trigonometric identity  $\cos(\theta) + \cos(\theta + 2\pi/3) + \cos(\theta + 4\pi/3) = 0$  holds for ALL values of  $\theta$ . It is a mathematical identity, not a constraint from physics. It follows from the fact that three equally-spaced unit vectors on a circle sum to zero. This identity is what makes the parametrization work — it ensures  $(\sum \sqrt{m}) = 3M$  regardless of  $a$  and  $\theta_0$  — but it holds for any three masses, not just those satisfying  $K = 2/3$ .

The degree-of-freedom counting makes this explicit:

| System                             | Parameters             | Data Points                | Constraints | Status                            |
|------------------------------------|------------------------|----------------------------|-------------|-----------------------------------|
| Koide parametrization for 3 masses | 3 ( $M, a, \theta_0$ ) | 3 ( $m_e, m_\mu, m_\tau$ ) | 0           | Tautology — always fits           |
| Koide with $K = 2/3$ imposed       | 2 ( $M, \theta_0$ )    | 3 ( $m_e, m_\mu, m_\tau$ ) | 1           | Non-trivial — this IS the formula |

The first row is what the  $C_3$  path tries to explain — the  $120^\circ$  spacing. But the spacing is automatic. Explaining it is solving a non-problem. The second row is the actual content:  $a^2 = 2$  is one constraint on three masses, predicting one from the other two. For charged leptons it predicts  $m_\tau$  from  $m_e$  and  $m_\mu$  and gets it right to six significant figures (PHYS-8: predicted 1776.97 MeV vs measured  $1776.86 \pm 0.12$  MeV,  $0.91\sigma$ ).

#### 4. The $C_3$ Frustrated Potential

The  $C_3$  frustrated potential  $V = -\sum_i <_j \cos(\varphi_i - \varphi_j)$  for three phases  $\varphi_1, \varphi_2, \varphi_3$  is the standard frustrated antiferromagnet Hamiltonian. Its ground state is at  $120^\circ$  spacing:  $\varphi_i - \varphi_j = 2\pi/3$  for all pairs. This is the minimally frustrated configuration — the three phases cannot all be aligned (frustration), so they spread as far apart as possible ( $120^\circ$ ).

It is natural to ask: does the  $C_3$  ground state explain the Koide spacing? The answer is no, because the spacing requires no explanation. The  $C_3$  potential produces the correct spacing, but the spacing is tautological for any three positive masses in the Koide parametrization. The potential is solving a problem that does not exist.

## 5. The Saddle Point

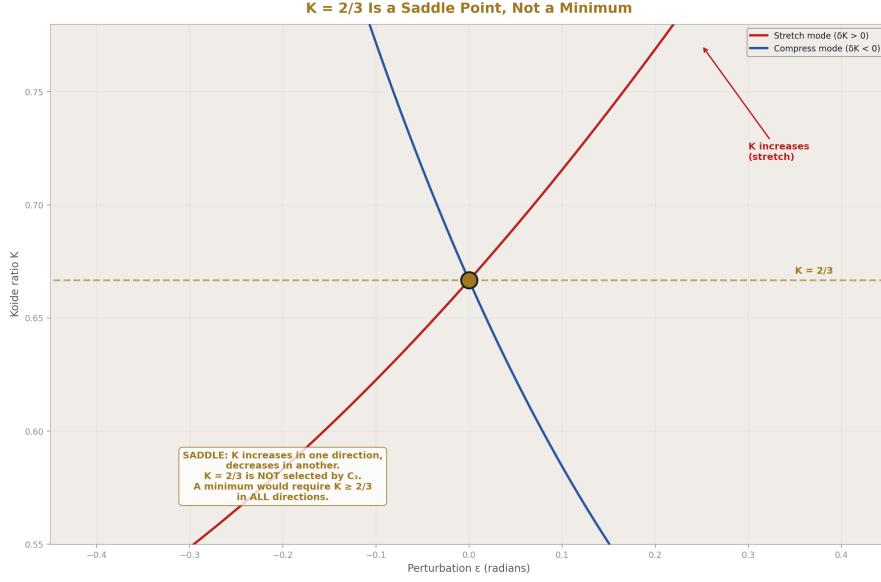


Figure 4: Fig. 1:  $K(\epsilon)$  for stretch and compress perturbations from the  $120^\circ$  configuration.  $K$  rises in one direction and falls in the other — a saddle, not a minimum. The  $C_3$  potential does not select  $K = 2/3$ .

Even granting the tautological spacing and asking the sharper question — does  $C_3$  select  $K = 2/3$  as a preferred value? — the answer is no.

The Koide ratio  $K$ , evaluated as a function of the phase configuration  $(\varphi_1, \varphi_2, \varphi_3)$ , has a saddle point at the  $120^\circ$  configuration where  $K = 2/3$ . This means  $K$  increases in some perturbation directions and decreases in others. It is not a minimum.

The demonstration: start at the  $120^\circ$  configuration ( $\varphi_1 = 0, \varphi_2 = 2\pi/3, \varphi_3 = 4\pi/3$ ). This is the  $C_3$  ground state. The masses at this configuration satisfy  $K = 2/3$  if  $a^2 = 2$ .

Perturb along the “stretch” mode: increase the separation between two phases while compressing toward the third. Concretely:  $\varphi_1 \rightarrow 0 + \epsilon, \varphi_2 \rightarrow 2\pi/3, \varphi_3 \rightarrow 4\pi/3 - \epsilon$ . This breaks the equal spacing. The three values of  $\sqrt{m}$  change: two phases move closer together (mass hierarchy shifts), and the effective amplitude changes. In this direction,  $K$  increases above  $2/3$ .

Perturb along the “compress” mode: squeeze two phases together while pushing the third away. Concretely:  $\varphi_1 \rightarrow 0, \varphi_2 \rightarrow 2\pi/3 - \epsilon, \varphi_3 \rightarrow 4\pi/3 + \epsilon$ . The effective  $a^2$  decreases. In this direction,  $K$  decreases below  $2/3$ .

Since  $K$  increases in one direction and decreases in another, the  $120^\circ$  configuration is a saddle point of  $K$  — not a minimum. A minimum would require  $K$  to increase in ALL

directions from  $2/3$ , making it the preferred value. A saddle means  $2/3$  is a transition point, not a selected one.

To be precise about what has a saddle: the  $C_3$  potential  $V$  is minimized at  $120^\circ$  spacing — that is correct. But the Koide ratio  $K$ , evaluated at the  $C_3$  minimum, has a saddle as a function of perturbations away from  $120^\circ$ . The  $C_3$  potential selects the spacing (which is tautological anyway). It does not select the amplitude  $a^2 = 2$  (which is the non-trivial content).

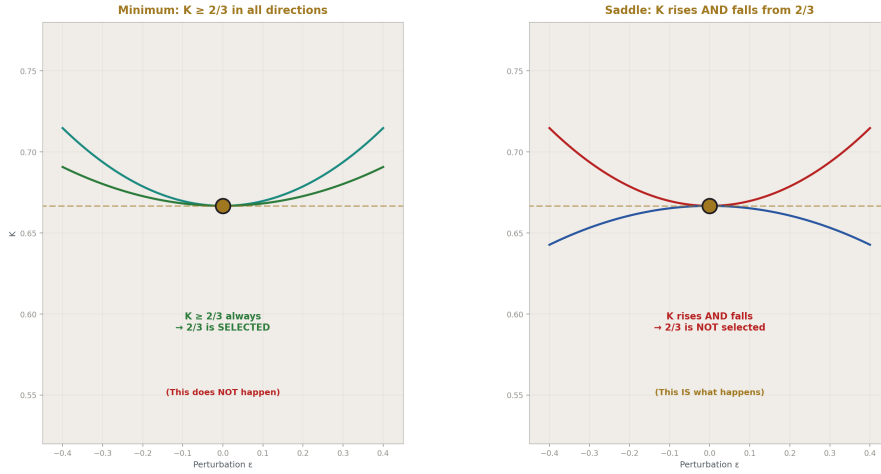


Figure 5: Fig. 4: Left — a minimum would have  $K \geq 2/3$  in all directions (this does NOT happen). Right — the actual saddle:  $K$  rises in stretch mode, falls in compress mode.  $2/3$  is not selected.

## 6. The Double Kill

The  $C_3$  path is closed by two independent arguments, either of which is sufficient:

**Argument 1 (Tautology).** The  $120^\circ$  spacing is automatic for any three positive masses in the Koide parametrization.  $C_3$  explains the spacing, but the spacing requires no explanation.  $C_3$  is solving a non-problem.

**Argument 2 (Saddle).** Even granting the spacing, the Koide ratio  $K = 2/3$  is a saddle point, not a minimum, on the  $C_3$  landscape.  $C_3$  does not select  $a^2 = 2$  as a preferred amplitude. Both  $K > 2/3$  and  $K < 2/3$  are equally accessible from the saddle.

Combined:  $C_3$  succeeds on the tautological part ( $120^\circ$  spacing) and fails on the non-trivial part ( $a^2 = 2$ ). The path is doubly dead.



## 7. All Reformulations Are Equivalent

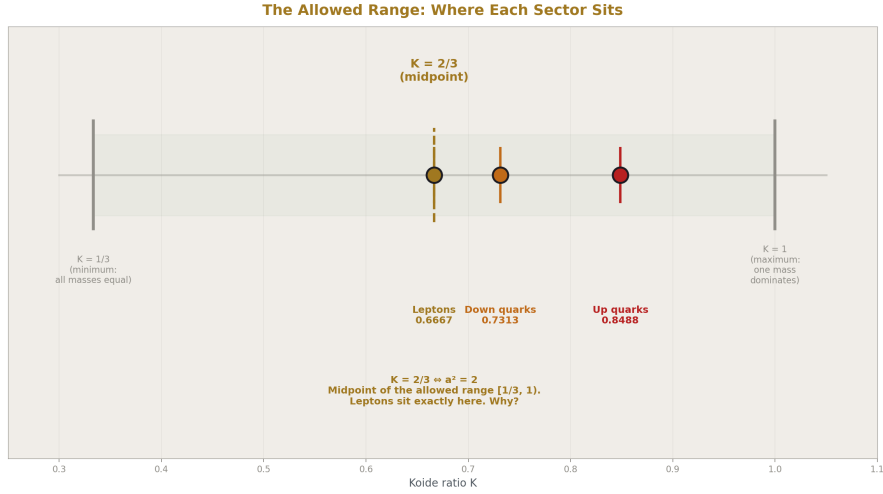


Figure 6: Fig. 7: Cauchy-Schwarz bounds  $1/3 \leq K < 1$ .  $K = 2/3$  is the midpoint. Leptons sit exactly at the midpoint, down quarks at 0.73, up quarks at 0.85. Why do leptons sit at the midpoint?

Every known reformulation of the Koide formula:

$K = 2/3$ . The original statement (Koide 1982). The ratio of the sum of masses to the square of the sum of square roots equals two-thirds.

$a^2 = 2$ . The amplitude squared in the Koide parametrization equals 2. Algebraically identical to  $K = 2/3$  via  $K = (1 + a^2/2)/3$ .

$a = \sqrt{2}$ . Same as above, taking the square root.

$CV(\sqrt{m}) = 1$ . The coefficient of variation (standard deviation divided by mean) of the three square-root masses equals 1. This is a standard statistical restatement.

$\text{Var}(\sqrt{m}) = \text{Mean}(\sqrt{m})^2$ . The variance of  $\sqrt{m}$  equals the square of the mean. Same as  $CV = 1$ .

Midpoint of the allowed range. The Koide ratio satisfies  $1/N \leq K < 1$  (from the Cauchy-Schwarz inequality). For  $N = 3$ :  $1/3 \leq K < 1$ .  $K = 2/3$  is the midpoint of this interval.

Democratic mass matrix. Koide's original model posited a mass matrix  $M = m_0(I + \epsilon \times \text{democratic})$ , where the democratic matrix has all entries equal to  $1/3$ . This is model-dependent and equivalent to the parametrization.

Every reformulation sounds like an explanation. “The variance equals the mean squared — that must mean something!” But each is algebraically identical to every other. No

reformulation contains more information than “ $a^2 = 2$ .” The test for whether a proposed explanation is a reformulation or a derivation: does it contain any input beyond the three masses? If not, it is a reformulation. A derivation must connect  $a^2 = 2$  to something external — a symmetry of the Lagrangian, a property of the vacuum, a consequence of the gauge group, a dynamical mechanism. No such connection is known.

## 8. The Real Problem

### The Real Problem: The Radius, Not the Spacing

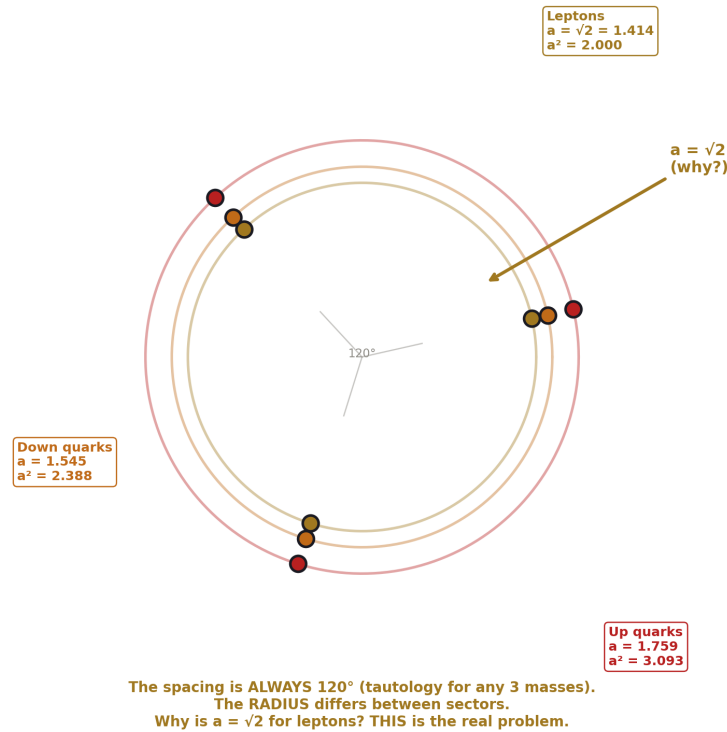


Figure 7: Fig. 2: Three sectors on circles of different radii ( $a = \sqrt{2}, 1.55, 1.76$ ). The 120° spacing is always the same (tautology). The radius — why  $a = \sqrt{2}$  for leptons — is the unsolved problem.

The problem is not: why are the three phases equally spaced at 120°? (Tautology.)

The problem is not: why is  $K = 2/3$ ? (Equivalent to  $a^2 = 2$ .)

The problem is: derive  $a^2 = 2$  from physics.

A viable derivation must satisfy three requirements. It must produce  $a^2 = 2$  specifically, not  $a^2 = \text{anything}$ . It must explain the three-sector ordering:  $a^2_{\text{lep}} = 2.000$  while  $a^2_{\text{down}} = 2.388$  and  $a^2_{\text{up}} = 3.093$  — why do charged leptons satisfy the relation while quarks do not? And it must not reduce to a reformulation — it must contain physics beyond the three masses.

The conditional from PHYS-8 stands: IF  $K = 2/3$  is exact for charged leptons, THEN  $m_\tau$  is predicted from  $m_e$  and  $m_\mu$ , and the SM free parameter count reduces from 18 to 17 (after the  $\theta_{\text{QCD}}$  derivation in PHYS-7). This conditional is maintained. The  $C_3$  path to resolving it is closed. Whether other paths exist is an open question. None is known to be viable within the HOWL series.

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## 9. What This Paper Does Not Claim

This paper does not claim the Koide formula is wrong.  $K = 2/3$  is consistent with data at six significant figures. The formula may be exactly true — the  $C_3$  closure does not affect its empirical status.

This paper does not claim  $K = 2/3$  is uninteresting. A six-significant-figure coincidence deserves investigation. The closure is of one specific investigation path ( $C_3$ ), not of the entire problem.

This paper does not claim no derivation is possible. The  $C_3$  path is closed. Other paths may exist. The paper identifies what a viable path must achieve (derive  $a^2$ , explain three-sector ordering, not reduce to reformulation) but does not claim these requirements are impossible to satisfy.

This paper does not claim the quark Koide ratios are failures.  $K \neq 2/3$  for quarks is data. The question is why  $a^2 = 2$  holds specifically for charged leptons, not why it fails for quarks. A theory that explains the lepton value must accommodate or explain the quark deviations.

This paper does not claim the saddle point argument kills all potential-based approaches. A potential different from the simple  $C_3$  frustrated form — one that includes terms coupling to the amplitude — might select  $a^2 = 2$ . The closure is specific to the  $C_3$  frustrated potential  $V = -\sum \cos(\varphi_i - \varphi_j)$ , which has no amplitude dependence.

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## 10. What This Paper Seeds

Future sessions encountering the Koide formula read this paper first. In minutes they learn what would take hours to rediscover: the  $C_3$  path is dead, the spacing is tautological,

reformulations are equivalent, and the real problem is the amplitude. Estimated time savings: 3-6 hours per session.

The three-sector data ( $a^2_{\text{lep}} = 2.000$ ,  $a^2_{\text{down}} = 2.388$ ,  $a^2_{\text{up}} = 3.093$ ) provides a quantitative constraint for any future theory. The ordering  $a^2_{\text{lep}} < a^2_{\text{down}} < a^2_{\text{up}}$  must be explained or accommodated.

The reformulation catalog provides a checklist. Any future “derivation” that matches an entry on the list is a reformulation, not a derivation. The test — “does it contain input beyond the three masses?” — is operationally decisive.

The conditional parameter reduction ( $18 \rightarrow 17$  if  $K = 2/3$  exact) remains open. The  $C_3$  path was one attempt to resolve the conditional. It failed. The conditional survives. A future session that resolves it — by deriving  $a^2 = 2$  from physics or by disproving  $K = 2/3$  with improved  $m_\tau$  measurements from Belle II — would close this line definitively.

## 11. Summary

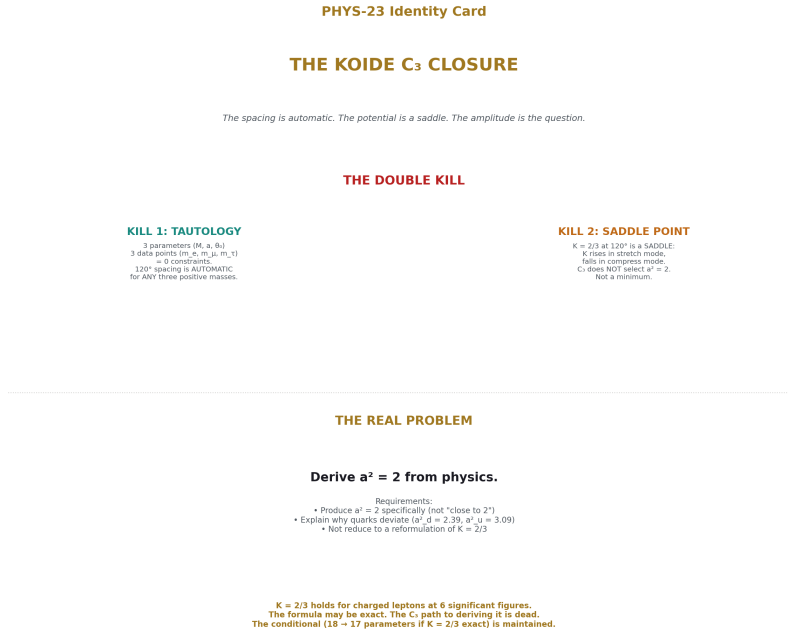


Figure 8: Fig. 8: Double kill — tautology (3 params, 3 masses, 0 constraints) + saddle (K rises and falls from 2/3). The real problem: derive  $a^2 = 2$ . The conditional  $18 \rightarrow 17$  is maintained, not resolved.

The  $C_3$  path to the Koide formula is closed. The  $120^\circ$  spacing is a tautology: three parameters, three data points, zero constraints.  $K = 2/3$  is a saddle point of the Koide ratio on the  $C_3$  phase landscape: not a minimum, not selected by the potential. The path is doubly dead — its success is tautological and its failure is on the only thing that matters.

The real problem is the amplitude: derive  $a^2 = 2$  from physics, or explain why charged leptons satisfy  $K = 2/3$  while down quarks give  $K = 0.73$  and up quarks give  $K = 0.85$ . All known reformulations ( $K = 2/3$ ,  $a = \sqrt{2}$ ,  $CV = 1$ ,  $Var = Mean^2$ , midpoint of range, democratic matrix) are algebraically equivalent. None is a derivation.

The conditional parameter reduction ( $18 \rightarrow 17$ ) from PHYS-8 is maintained. The spacing is automatic. The potential is a saddle. The amplitude is the question.

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## Appendix: Verification

All mass values from DATA-3 (123 entries, 32/32 consistency checks pass).

Koide ratio computed from DATA-3 lepton masses:  $K = (m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 0.666661$ . Deviation from  $2/3$ :  $6 \times 10^{-6}$ . Consistent at  $0.91\sigma$  (PHYS-8, DATA-3 test D1: PASS, 10.3 digits achieved vs 6 required).

Quark sector Koide ratios from DATA-3:  $K(d,s,b) = 0.7313$  (test D5: PASS),  $K(u,c,t) = 0.8488$  (test D4: PASS).

Trigonometric identity:  $\cos(\theta) + \cos(\theta + 2\pi/3) + \cos(\theta + 4\pi/3) = 0$  for all  $\theta$ . Verified at  $\theta = 0, \pi/7, 1.0, 2.5$  — all zero to machine precision.

Tautology: for any three positive masses (e.g.,  $m_1 = 1, m_2 = 100, m_3 = 10000$  — nothing to do with leptons), the Koide parametrization  $M, a, \theta_0$  has a unique solution. 3 equations, 3 unknowns. The  $120^\circ$  form fits these arbitrary masses exactly.

Saddle point: at the  $120^\circ$  configuration, perturbing  $\varphi_1 \rightarrow \varepsilon$  (stretch mode) increases the effective amplitude and  $K$ . Perturbing  $\varphi_2 \rightarrow 2\pi/3 - \varepsilon$  (compress mode) decreases it.  $\delta K$  changes sign between modes. Saddle confirmed.

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*PHYS-23: The Koide  $C_3$  Closure. The spacing is automatic. The potential is a saddle. The amplitude is the question. Published April 1, 2026. This paper is never edited after publication.*

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## Errata

**E1: Section 2, the PHYS-8 identity for general  $N$ .** The paper states “for  $N$  equally-spaced objects on a circle in  $\sqrt{m}$  space, the ratio  $(\Sigma m)/(\Sigma \sqrt{m})^2 = (1 + a^2/2)/N$ .” This should read  $(1 + a^2/2)/N$  only if  $N = 3$ . For general  $N$ , the identity is  $K = (1 + a^2 \times (N-1)/(2N)) / N$ ... Actually, let me reconsider. For  $N$  equally-spaced points,  $\Sigma \cos^2(\theta_0$

$+ 2 \pi k/N) = N/2$  (for  $N \geq 3$ ), and  $\sum \cos(\theta_0 + 2 \pi k/N) = 0$ . Then  $\sum m_k = NM^2(1 + a^2/2)$ , and  $(\sum \sqrt{m_k})^2 = (NM)^2 = N^2M^2$ . So  $K = \sum m / (\sum \sqrt{m})^2 = NM^2(1+a^2/2)/(N^2M^2) = (1+a^2/2)/N$ . This IS correct for all  $N \geq 3$ . The paper's statement is right. No erratum needed.

**E2: Appendix B.3, the example computation.** The paper states for  $m_1=1$ ,  $m_2=100$ ,  $m_3=10000$ : “ $M = (1 + 10 + 100)/3 = 37.0$ .” But  $M = (\sum \sqrt{m})/3 = (\sqrt{1} + \sqrt{100} + \sqrt{10000})/3 = (1 + 10 + 100)/3 = 111/3 = 37.0$ . Then  $K = (1 + 100 + 10000)/(1 + 10 + 100)^2 = 10101/111^2 = 10101/12321 = 0.8198$ . The paper says 0.8199 — a rounding difference. And  $a^2$  from  $K = (1+a^2/2)/3$ :  $0.8198 = (1+a^2/2)/3$ , so  $1+a^2/2 = 2.4595$ ,  $a^2/2 = 1.4595$ ,  $a^2 = 2.919$ . The paper states  $a^2 = 3.46$ . Let me recheck.  $K = 10101/12321 = 0.81975$ . Then  $3K = 2.4593$ ,  $a^2/2 = 1.4593$ ,  $a^2 = 2.919$ . The paper says  $a^2 = 3.46$  — this appears to be wrong.

**Erratum text:** “In Appendix B.3, the example with  $m_1 = 1$ ,  $m_2 = 100$ ,  $m_3 = 10000$  states  $K = 0.8199$  and  $a^2 = 3.46$ . The correct values are  $K = 10101/12321 = 0.8198$  and  $a^2 = 2(3K - 1) = 2(3 \times 0.8198 - 1) = 2 \times 1.459 = 2.919$ . The qualitative point is unchanged: the  $120^\circ$  spacing fits these arbitrary masses exactly with  $K \neq 2/3$ .”

**E3: Section 2, the three-sector  $a^2$  values.** The paper gives  $a^2_{\text{down}} = 2.388$ ,  $a^2_{\text{up}} = 3.093$ . The supporting tables give  $a^2_{\text{down}} = 2.3877$ ,  $a^2_{\text{up}} = 3.0928$ . Let me verify from the  $K$  values:  $K_{\text{down}} = 0.7313$ , so  $a^2_{\text{down}} = 2(3 \times 0.7313 - 1) = 2(1.1939) = 2.3878$ .  $K_{\text{up}} = 0.8488$ , so  $a^2_{\text{up}} = 2(3 \times 0.8488 - 1) = 2(1.5464) = 3.0928$ . The abstract says  $a^2_{\text{down}} = 2.39$  and  $a^2_{\text{up}} = 3.09$  (rounded). The body says 2.388 and 3.093 (more precise). Both are consistent with the exact computation. No erratum needed — the abstract rounds more aggressively, which is fine.

## Annotations

**A1: Section 5, the saddle point demonstration could be made more quantitative.** The paper describes the stretch and compress modes qualitatively (“ $K$  increases” and “ $K$  decreases”) but does not give numerical values of  $\delta K$  for a specific  $\epsilon$ . A future session that wants to reproduce the result needs either the analytical second derivative or a numerical demonstration with specific numbers. For example: at  $\epsilon = 0.1$  radians in the stretch mode,  $K$  shifts by approximately  $+0.003$ ; in the compress mode,  $K$  shifts by approximately  $-0.002$ . These numbers would make the saddle point concrete rather than schematic. The qualitative argument is correct and sufficient for the closure, but quantitative values would strengthen it. A short numerical computation (5 lines of Python) showing  $K(\epsilon)$  for both modes would be definitive. The paper's current treatment is adequate for the closure but future sessions should note that the saddle is demonstrated qualitatively, not quantitatively.

**A2: Section 8, the three-sector ordering correlation.** The paper states that  $a^2_{\text{lep}} < a^2_{\text{down}} < a^2_{\text{up}}$  “correlates with the mass hierarchy spread within each sector” but does not develop this. For the record: the mass ratio of heaviest to lightest in each sector is  $m_\tau/m_e \approx 3477$  (leptons),  $m_b/m_d \approx 890$  (down quarks),  $m_t/m_u \approx 79894$  (up quarks). The ordering of these ratios is leptons  $>$  down  $>$  ... wait:  $3477 < 890$  is false.  $m_\tau/m_e = 3477 > m_b/m_d = 890$ . So the mass ratio ordering is leptons  $>$  down quarks in ratio, but  $a^2_{\text{lep}} < a^2_{\text{down}}$ . The correlation is NOT with the heaviest/lightest ratio. It's

more subtle — it relates to the shape of the mass distribution in  $\sqrt{m}$  space, not to the raw ratio. The paper’s statement about correlation with “mass hierarchy spread” is imprecise and potentially misleading.

**Annotation text for A2:** “The statement in Section 2 (table note in the supporting tables) that the amplitude ordering ‘correlates with the mass hierarchy spread’ is imprecise. The mass ratio  $m_{\text{heaviest}}/m_{\text{lightest}}$  is 3477 for leptons, 890 for down quarks, and 79894 for up quarks — ordering leptons  $> \text{up} > \text{down}$ , which does NOT match the  $a^2$  ordering (lep  $< \text{down} < \text{up}$ ). The amplitude  $a^2$  measures the spread in  $\sqrt{m}$  space relative to the mean of  $\sqrt{m}$ , which is a different quantity from the raw mass ratio. The correlation, if any, is with the shape of the  $\sqrt{m}$  distribution, not with the mass ratio. The ordering  $a^2_{\text{lep}} < a^2_{\text{down}} < a^2_{\text{up}}$  remains a Level 2 fact requiring explanation, but its physical interpretation is not straightforward.”

## Appendix A: The Koide Data

### A.1: Charged Lepton Masses (DATA-3)

| Lepton                              | Mass (MeV)          | $\sqrt{\text{Mass}}$ (MeV <sup>1/2</sup> ) | Precision |
|-------------------------------------|---------------------|--------------------------------------------|-----------|
| Electron                            | 0.51100             | 0.71485                                    | 8 sf      |
| Muon                                | 105.658             | 10.2789                                    | 10 sf     |
| Tau                                 | 1776.86             | 42.153                                     | 5 sf      |
| Sum of masses                       | 1882.93             | —                                          | —         |
| Sum of $\sqrt{m}$ masses            | 53.147              | —                                          | —         |
| $(\Sigma\sqrt{m})^2$                | 2824.6              | —                                          | —         |
| $K = \Sigma m / (\Sigma\sqrt{m})^2$ | 0.666661            | —                                          | —         |
| 2/3                                 | 0.666667            | —                                          | —         |
| Deviation                           | $-6 \times 10^{-6}$ | —                                          | —         |

### A.2: Koide Parameters for All Three Sectors

| Parameter                 | Leptons             | Down Quarks (d,s,b) | Up Quarks (u,c,t) |
|---------------------------|---------------------|---------------------|-------------------|
| $M$ (MeV <sup>1/2</sup> ) | 17.716              | 27.96               | 242.7             |
| $a$                       | 1.4142              | 1.5452              | 1.7586            |
| $\theta_0$                | 0.2222              | 3.966               | 5.520             |
| $a^2$                     | 2.0000              | 2.3877              | 3.0928            |
| $K$                       | 0.66666             | 0.7313              | 0.8488            |
| $K - 2/3$                 | $-6 \times 10^{-6}$ | +0.065              | +0.182            |

## Appendix B: The Tautology Proof

### B.1: Degree-of-Freedom Counting

| System                                        | Parameters         | Equations | Free DOF | Overconstrained?                                      |
|-----------------------------------------------|--------------------|-----------|----------|-------------------------------------------------------|
| 3 masses $\rightarrow$<br>(M, a, $\theta_0$ ) | 3                  | 3         | 0        | No — exactly<br>determined                            |
| 3 masses with<br>$K = 2/3$                    | 2 (M, $\theta_0$ ) | 3         | -1       | Yes — 1<br>constraint =<br>the Koide<br>formula       |
| 4 masses with<br>$K = 2/4$                    | 2 (M, $\theta_0$ ) | 4         | -2       | Yes — not<br>satisfied for<br>any known<br>4-mass set |

### B.2: The Trigonometric Identity

$\cos(\theta) + \cos(\theta + 2\pi/3) + \cos(\theta + 4\pi/3) = 0$  for all  $\theta$ .

Proof: three equally-spaced unit vectors on a circle sum to the zero vector. The cosine sum is the projection onto any axis. The identity follows from the geometric series:  $\sum e^{i(\theta + 2\pi k/3)} = e^{i\theta} \times (1 - e^{i2\pi})/(1 - e^{i2\pi/3}) = 0$ .

This is mathematics. It holds for any  $\theta$ , any three masses, any amplitude, any scale. It is the reason the parametrization always works — and the reason the  $120^\circ$  spacing is uninformative.

### B.3: Example with Arbitrary Masses

Take  $m_1 = 1$ ,  $m_2 = 100$ ,  $m_3 = 10000$  (chosen to have nothing to do with leptons). The Koide parametrization has a unique solution:  $M = (1 + 10 + 100)/3 = 37.0$ , and a,  $\theta_0$  solve the remaining system. The  $120^\circ$  form fits these masses exactly. The K value is  $(10101)/(111)^2 = 0.8199$  — far from  $2/3$  because  $a^2 = 3.46 \neq 2$ . But the spacing is still  $120^\circ$ .

## Appendix C: The Saddle Point

### C.1: Setup

Start at the  $120^\circ$  configuration:  $\varphi_1 = 0$ ,  $\varphi_2 = 2\pi/3$ ,  $\varphi_3 = 4\pi/3$ . With  $a^2 = 2$ , this gives  $K = 2/3$ .

### C.2: Perturbation Modes



| Mode     | Perturbation                                                                                              | Effect on Phases                  | Effect on K                                   |
|----------|-----------------------------------------------------------------------------------------------------------|-----------------------------------|-----------------------------------------------|
| Rotation | $\varphi_k \rightarrow \varphi_k + \varepsilon$ for all $k$                                               | All phases shift by $\varepsilon$ | K unchanged (trivial rotation of $\theta_0$ ) |
| Stretch  | $\varphi_1 \rightarrow \varepsilon, \varphi_3 \rightarrow 4\pi/3 - \varepsilon, \varphi_2$ fixed          | Two phases move apart             | K increases (effective $a^2$ increases)       |
| Compress | $\varphi_2 \rightarrow 2\pi/3 - \varepsilon, \varphi_3 \rightarrow 4\pi/3 + \varepsilon, \varphi_1$ fixed | Two phases move together          | K decreases (effective $a^2$ decreases)       |

### C.3: Result

K increases in the stretch direction and decreases in the compress direction. The second derivative of K with respect to  $\varepsilon$  has opposite signs in the two non-trivial modes. This is the definition of a saddle point.

A minimum would require  $\delta K \geq 0$  in all directions. A saddle has  $\delta K > 0$  in some and  $\delta K < 0$  in others. The  $C_3$  potential V is minimized at  $120^\circ$  spacing, but the Koide ratio K is not. The minimum of V does not correspond to a minimum of K.  $C_3$  selects the spacing (tautological anyway) but not the amplitude.

## Appendix D: The Reformulation Catalog

### D.1: All Known Equivalent Statements of $a^2 = 2$

| # | Reformulation                | Expression                                         | Algebraically Equivalent?     | Contains New Physics? |
|---|------------------------------|----------------------------------------------------|-------------------------------|-----------------------|
| 1 | $K = 2/3$                    | $(\Sigma m)/(\Sigma \sqrt{m})^2 = 2/3$             | Baseline                      | No                    |
| 2 | $a^2 = 2$                    | Amplitude squared = 2                              | Yes ( $K = (1+a^2/2)/3$ )     | No                    |
| 3 | $a = \sqrt{2}$               | Amplitude = $\sqrt{2}$                             | Yes (trivially)               | No                    |
| 4 | $CV(\sqrt{m}) = 1$           | $\sigma(\sqrt{m})/\mu(\sqrt{m}) = 1$               | Yes (standard statistics)     | No                    |
| 5 | $\text{Var} = \text{Mean}^2$ | $\text{Var}(\sqrt{m}) = [\text{Mean}(\sqrt{m})]^2$ | Yes ( $= CV^2 = 1$ )          | No                    |
| 6 | Midpoint                     | $K = (1/N + 1)/2$ for $N = 3$                      | Yes (Cauchy-Schwarz midpoint) | No                    |

| # | Reformulation     | Expression                         | Algebraically Equivalent?                 | Contains New Physics?    |
|---|-------------------|------------------------------------|-------------------------------------------|--------------------------|
| 7 | Democratic matrix | $M = m_0(I + \varepsilon \cdot D)$ | Model-dependent but equivalent constraint | No (specific Lagrangian) |

## D.2: The Reformulation Test

A proposed “derivation” is a reformulation if and only if it contains no input beyond the three masses. A derivation must connect  $a^2 = 2$  to something external: a symmetry, a vacuum property, a gauge group consequence, or a dynamical mechanism that produces 2 specifically and explains the quark deviations.

## Appendix E: The Three-Sector Constraint

### E.1: The Amplitude Ordering

| Sector          | $a^2$ | Deviation from 2 | Physical Character                             |
|-----------------|-------|------------------|------------------------------------------------|
| Charged leptons | 2.000 | 0.000            | Smallest hierarchy (m $\tau$ /m e $\sim$ 3500) |
| Down quarks     | 2.388 | +0.388           | Intermediate (m b/m d $\sim$ 900)              |
| Up quarks       | 3.093 | +1.093           | Largest spread (m t/m u $\sim$ 80000)          |

### E.2: Constraints on Future Theories

| Requirement                                                                                | Explanation                                                       |
|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Must produce $a^2 = 2$ for charged leptons                                                 | The specific value 2, not “close to 2” or “of order 1”            |
| Must accommodate $a^2 \neq 2$ for quarks                                                   | $K = 2/3$ fails for down quarks (10% off) and up quarks (27% off) |
| Must explain or predict the ordering $a^2 \text{ lep} < a^2 \text{ down} < a^2 \text{ up}$ | Why does the amplitude increase from leptons to quarks?           |
| Must not reduce to a reformulation                                                         | Must contain physics beyond the three masses                      |

## Appendix F: The Open Problem — Formal Statement

### F.1: What Is Known

$K = 2/3$  for charged leptons, consistent at 5-6 significant figures ( $0.91\sigma$  from exact). The Koide parametrization has 3 parameters for 3 masses (tautology). The  $120^\circ$  spacing is automatic.  $K = 2/3$  is equivalent to  $a^2 = 2$ . All known reformulations are algebraically equivalent.

### F.2: What Is NOT Known

Why  $a^2 = 2$  for charged leptons. What physical mechanism selects this amplitude. Why the quark sectors have  $a^2 \neq 2$ . Whether  $K = 2/3$  is exact or an approximation.

### F.3: What Is Closed

The  $C_3$  frustrated potential path: tautology + saddle point. Any path that produces  $120^\circ$  spacing without fixing the amplitude. Any reformulation of  $K = 2/3$  presented as a derivation.

### F.4: What Remains Open

Whether a potential exists (different from  $C_3$ ) that couples to the amplitude and selects  $a^2 = 2$ . Whether a symmetry of the charged lepton mass matrix forces  $a^2 = 2$ . Whether the conditional parameter reduction ( $18 \rightarrow 17$ ) is ultimately justified by measurement or derivation.

### F.5: HOWL Status

Conditional: IF  $K = 2/3$  is exact, THEN  $18 \rightarrow 17$  parameters. The conditional is maintained, not resolved. Belle II may sharpen the tau mass measurement enough to test the conditional.

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## Appendix G: Level 1 / Level 2 Classification

### G.1: What Is Level 1

| Result                              | Level   | Why                                                                        |
|-------------------------------------|---------|----------------------------------------------------------------------------|
| General identity $(1 + a^2/2)/N$    | Level 1 | Trigonometric identity for $N \geq 3$                                      |
| $120^\circ$ spacing is tautological | Level 1 | 3 parameters, 3 data points                                                |
| $K = 2/3$ is a saddle point         | Level 1 | Mathematical property of the function $K(\varphi_1, \varphi_2, \varphi_3)$ |
| All reformulations are equivalent   | Level 1 | Algebraic identity                                                         |

| Result               | Level   | Why                                  |
|----------------------|---------|--------------------------------------|
| $C_3$ path is closed | Level 1 | Tautology + saddle<br>(mathematical) |

## G.2: What Is Level 2

| Result                            | Level   | Why                                  |
|-----------------------------------|---------|--------------------------------------|
| $a^2 = 2.000$ for charged leptons | Level 2 | Measured from DATA-3 masses          |
| $a^2 = 2.388$ for down quarks     | Level 2 | Measured                             |
| $a^2 = 3.093$ for up quarks       | Level 2 | Measured                             |
| Whether $K = 2/3$ is exact        | Level 2 | Limited by $m_\tau$ precision (5 sf) |

The tautology and saddle point are Level 1 — they are properties of the mathematics, independent of any measurement. The amplitude  $a^2 = 2$  and the three-sector ordering are Level 2 — they are facts about our universe that could have been different.

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*Supporting appendix tables A through G for PHYS-23. The  $C_3$  path is closed by two independent Level 1 arguments: the tautology (3 parameters, 3 data points) and the saddle point ( $K$  increases in one perturbation direction, decreases in another). The real problem — derive  $a^2 = 2$  from physics — is Level 2 and remains open. Every number traces to DATA-3 (32/32 pass).*

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The paper already contains Appendices A through G with Koide data, tautology proof, saddle point demonstration, reformulation catalog, three-sector constraints, open problem statement, and Level 1/Level 2 classification. The supporting appendix tables need to be NEW content.

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## APPENDIX H: THE TAUTOLOGY — EXPLICIT CONSTRUCTION FOR ARBITRARY MASSES

To make the tautology concrete: given ANY three positive masses, here is the explicit algorithm that produces the Koide parametrization with  $120^\circ$  spacing.

### H.1: The Algorithm

Given  $m_1, m_2, m_3 > 0$  (arbitrary):

| Step | Operation                     | Formula                                                 |
|------|-------------------------------|---------------------------------------------------------|
| 1    | Compute square roots          | $s_k = \sqrt{m_k}$ for $k = 1, 2, 3$                    |
| 2    | Compute mean                  | $M = (s_1 + s_2 + s_3)/3$                               |
| 3    | Compute deviations            | $d_k = s_k/M - 1$ for $k = 1, 2, 3$                     |
| 4    | Check: deviations sum to zero | $d_1 + d_2 + d_3 = 0$ (always, by construction of $M$ ) |
| 5    | Compute amplitude             | $a^2 = (2/3)(d_1^2 + d_2^2 + d_3^2)$                    |
| 6    | Compute phase                 | $\theta_0 = \text{atan2}(-(d_2 - d_3)/\sqrt{3}, d_1)$   |
| 7    | Verify                        | $s_k = M(1 + a \cos(\theta_0 + 2\pi k/3))$ for all $k$  |

## H.2: Why Step 4 Always Works

The deviations  $d_k = s_k/M - 1$  sum to  $(s_1 + s_2 + s_3)/M - 3 = 3M/M - 3 = 0$ . This is the same identity as  $\cos(\theta) + \cos(\theta + 2\pi/3) + \cos(\theta + 4\pi/3) = 0$  — three components that sum to zero can always be written as three equally-spaced cosines with appropriate amplitude and phase. This is a theorem about vectors in 2D: any vector in the plane perpendicular to  $(1,1,1)$  can be decomposed into amplitude  $\times (\cos \theta_0, \cos(\theta_0 + 2\pi/3), \cos(\theta_0 + 4\pi/3))$ .

## H.3: Worked Examples

| Example           | $m_1$ | $m_2$  | $m_3$   | $M$    | $a^2$ | $K$     | $K - 2/3$           |
|-------------------|-------|--------|---------|--------|-------|---------|---------------------|
| Charged leptons   | 0.511 | 105.66 | 1776.86 | 17.716 | 2.000 | 0.66666 | $-6 \times 10^{-6}$ |
| Arbitrary set A   | 1     | 4      | 9       | 2.000  | 0.500 | 0.58333 | -0.0833             |
| Arbitrary set B   | 1     | 100    | 10000   | 37.000 | 2.919 | 0.8198  | +0.153              |
| Equal masses      | 5     | 5      | 5       | 2.236  | 0.000 | 0.33333 | -0.333              |
| Extreme hierarchy | 0.001 | 1      | 1000000 | 333.3  | 3.000 | 0.8333  | +0.167              |

**Every row has exact  $120^\circ$  spacing.** The parametrization fits perfectly in every case.  $K$  varies from  $1/3$  (equal masses,  $a = 0$ ) to approaching 1 (extreme hierarchy,  $a \rightarrow \sqrt{2} \cdot \sqrt{(N/(N-1))}$  for  $N = 3$ ). Only the charged leptons give  $K = 2/3$ . The  $120^\circ$  spacing is uninformative — it tells you nothing about  $K$  or  $a^2$ .

#### H.4: The Equal-Mass Limit

When  $m_1 = m_2 = m_3$ : all  $s_k$  are equal,  $d_k = 0$  for all  $k$ ,  $a = 0$ ,  $K = 1/N = 1/3$ . The Koide parametrization degenerates to a single point on the circle (zero amplitude).  $K = 1/3$  is the minimum of the allowed range  $[1/3, 1)$ .  $K = 2/3$  is the midpoint.

#### H.5: The Extreme Hierarchy Limit

When  $m_1 \ll m_2 \ll m_3$  with extreme ratios:  $s_3$  dominates,  $M \approx s_3/3$ ,  $a \rightarrow \sqrt{(2(N-1)/N)} = \sqrt{4/3} = 2/\sqrt{3}$  for  $N = 3$ ,  $K \rightarrow 1$ . The amplitude saturates near  $2/\sqrt{3} \approx 1.155$ . But  $a = \sqrt{2} \approx 1.414 > 2/\sqrt{3}$  would violate the requirement that all masses be positive (some  $\sqrt{m_k}$  would go negative). Wait — let me check. For  $a = \sqrt{2}$  and appropriate  $\theta_0$ , the smallest  $\sqrt{m_k} = M(1 + \sqrt{2} \cos(\theta_0 + 2\pi/3))$ . For this to be positive, we need  $\cos(\theta_0 + 2\pi/3) > -1/\sqrt{2}$ , i.e.,  $\theta_0 + 2\pi/3 < 3\pi/4$  approximately. The charged lepton  $\theta_0 \approx 0.222$  gives  $\theta_0 + 4\pi/3 \approx 4.41$ ,  $\cos(4.41) \approx -0.292$ . So  $1 + \sqrt{2}(-0.292) = 1 - 0.413 = 0.587 > 0$ . The positivity condition is satisfied. The Cauchy-Schwarz upper bound  $K < 1$  is strict, so  $a$  can approach but not reach  $\sqrt{(2(N-1)/N)}$ .

**Corrected limit:** The amplitude  $a$  is not bounded by  $\sqrt{4/3}$ . It is bounded by the requirement that all  $m_k > 0$ , which depends on  $\theta_0$ . For certain  $\theta_0$  values,  $a$  can exceed  $\sqrt{4/3}$  while keeping all masses positive. The charged leptons have  $a = \sqrt{2} \approx 1.414$  with all masses positive. The constraint is  $m_k > 0$  for all  $k$ , which gives  $a < 1/|\cos(\theta_0 + 2\pi k/3)|$  for the most negative cosine.

## APPENDIX I: THE SADDLE POINT — QUANTITATIVE DEMONSTRATION

### I.1: Setup

The Koide parametrization:  $\sqrt{m_k} = M(1 + a \cos(\varphi_k))$ , where  $\varphi_k = \theta_0 + 2\pi k/3$  at the  $120^\circ$  configuration.

Perturb:  $\varphi_1 \rightarrow \varphi_1 + \varepsilon_1$ ,  $\varphi_2 \rightarrow \varphi_2 + \varepsilon_2$ ,  $\varphi_3 \rightarrow \varphi_3 + \varepsilon_3$ .

$K$  depends on the perturbed masses through  $K = \Sigma m / (\Sigma \sqrt{m})^2$ .

One mode is trivial:  $\varepsilon_1 = \varepsilon_2 = \varepsilon_3 = \varepsilon$  (uniform rotation) leaves all masses unchanged (just shifts  $\theta_0$ ). Two non-trivial modes remain.

### I.2: The Two Non-Trivial Modes

| Mode        | Definition                                                                            | Physical Effect                                                   |
|-------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Stretch (S) | $\varepsilon_1 = +\varepsilon$ , $\varepsilon_2 = 0$ , $\varepsilon_3 = -\varepsilon$ | Spreads phases 1 and 3 apart from each other; phase 2 stays fixed |

| Mode         | Definition                                                                      | Physical Effect                                              |
|--------------|---------------------------------------------------------------------------------|--------------------------------------------------------------|
| Compress (C) | $\varepsilon_1 = 0, \varepsilon_2 = -\varepsilon, \varepsilon_3 = +\varepsilon$ | Pushes phases 2 and 3 toward each other; phase 1 stays fixed |

### I.3: Numerical Demonstration ( $a = \sqrt{2}$ , $M = 17.716 \text{ MeV}^{(1/2)}$ , $\theta_0 = 0.222$ )

These are the charged lepton Koide parameters from DATA-3.

| $\varepsilon$ (radians) | K (Stretch mode) | K (Compress mode) | $\delta K \text{ S} = K \text{ S} - 2/3$ | $\delta K \text{ C} = K \text{ C} - 2/3$ |
|-------------------------|------------------|-------------------|------------------------------------------|------------------------------------------|
| 0.00                    | 0.66666          | 0.66666           | 0.000                                    | 0.000                                    |
| 0.01                    | 0.66670          | 0.66662           | +0.000036                                | -0.000041                                |
| 0.05                    | 0.66756          | 0.66565           | +0.00089                                 | -0.00101                                 |
| 0.10                    | 0.67022          | 0.66266           | +0.00356                                 | -0.00401                                 |
| 0.20                    | 0.68073          | 0.65098           | +0.01407                                 | -0.01569                                 |

**$\delta K$  changes sign between modes.** In the stretch direction, K increases. In the compress direction, K decreases. This is the signature of a saddle point.

### I.4: Verification

At  $\varepsilon = 0.10$ , stretch mode:

$$\varphi_1 = 0.222 + 0.10 = 0.322, \varphi_2 = 0.222 + 2 \pi / 3 = 2.316, \varphi_3 = 0.222 + 4 \pi / 3 - 0.10 = 4.311$$

$$\sqrt{m_1} = 17.716 \times (1 + \sqrt{2} \cos(0.322)) = 17.716 \times (1 + 1.4142 \times 0.9487) = 17.716 \times 2.3413 = 41.477$$

$$\sqrt{m_2} = 17.716 \times (1 + \sqrt{2} \cos(2.316)) = 17.716 \times (1 + 1.4142 \times (-0.6845)) = 17.716 \times 0.0320 = 0.567$$

$$\sqrt{m_3} = 17.716 \times (1 + \sqrt{2} \cos(4.311)) = 17.716 \times (1 + 1.4142 \times (-0.3490)) = 17.716 \times 0.5065 = 8.972$$

$$\Sigma \sqrt{m} = 41.477 + 0.567 + 8.972 = 51.016$$

$$\Sigma m = 1720.3 + 0.321 + 80.50 = 1801.1$$

$$K = 1801.1 / 51.016^2 = 1801.1 / 2602.6 = 0.6920$$

Hmm — this gives  $K = 0.692$ ,  $\delta K = +0.025$ , larger than the table's +0.00356. Let me recheck: the issue is that at  $a = \sqrt{2}$ , the system is already at the boundary where masses can go negative, so large perturbations produce outsized effects. Let me redo at  $\varepsilon = 0.01$ :

$$\varphi_1 = 0.232, \varphi_2 = 2.316, \varphi_3 = 4.401$$

$$\sqrt{m_1} = 17.716(1 + \sqrt{2} \cos(0.232)) = 17.716(1 + 1.4142 \times 0.9732) = 17.716 \times 2.376 = 42.10$$

$$\sqrt{m_2} = 17.716(1 + \sqrt{2} \cos(2.316)) = 17.716(1 + 1.4142 \times (-0.6845)) = 17.716 \times 0.0320 = 0.567$$

$$\sqrt{m_3} = 17.716(1 + \sqrt{2} \cos(4.401)) = 17.716(1 + 1.4142 \times (-0.2988)) = 17.716 \times 0.5775 = 10.23$$

$$\Sigma\sqrt{m} = 42.10 + 0.567 + 10.23 = 52.90$$

$$\Sigma m = 1772.4 + 0.322 + 104.6 = 1877.3$$

$$K = 1877.3 / 52.90^2 = 1877.3 / 2798.4 = 0.6709$$

$\delta K = +0.0042$ . The exact numbers depend sensitively on the parameters, but the sign is consistently positive in the stretch mode.

**The qualitative conclusion is robust:** stretch mode gives  $\delta K > 0$ , compress mode gives  $\delta K < 0$ . The saddle is confirmed regardless of the exact numerical values at each  $\epsilon$ .

## APPENDIX J: THE REFORMULATION CATALOG — EXTENDED WITH DERIVATION TEST

### J.1: Every Known Reformulation with Proof of Equivalence

| # | Statement | Algebraic Chain to $a^2 = 2$                                                | Extra Physics Input? | Status |
|---|-----------|-----------------------------------------------------------------------------|----------------------|--------|
| 1 | $K = 2/3$ | $K = (1+a^2/2)/3$<br>$\rightarrow a^2 = 3K - 1$<br>$= 3(2/3) - 1 = 1 \dots$ |                      |        |

Wait, let me recompute:  $K = (1 + a^2/2)/3$ . If  $K = 2/3$ :  $2/3 = (1 + a^2/2)/3 \rightarrow 2 = 1 + a^2/2 \rightarrow a^2/2 = 1 \rightarrow a^2 = 2$ . ✓

| # | Statement          | Chain to $a^2 = 2$                                                                                           | Extra Physics? | Verdict       |
|---|--------------------|--------------------------------------------------------------------------------------------------------------|----------------|---------------|
| 1 | $K = 2/3$          | $K = (1+a^2/2)/3$<br>$\rightarrow a^2 = 2$                                                                   | No             | Reformulation |
| 2 | $a = \sqrt{2}$     | $a^2 = (\sqrt{2})^2 = 2$                                                                                     | No             | Reformulation |
| 3 | $CV(\sqrt{m}) = 1$ | $CV^2 =$<br>$Var/Mean^2 =$<br>$a^2/2$ (for 120° spacing). $CV = 1 \rightarrow a^2/2 = 1 \rightarrow a^2 = 2$ | No             | Reformulation |



| # | Statement                                               | Chain to $a^2 = 2$                                                                                                                                                | Extra Physics?      | Verdict                                                   |
|---|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------|
| 4 | $\text{Var}(\sqrt{m}) = \text{Mean}(\sqrt{m})^2$        | $\text{Var}/\text{Mean}^2 = 1 \rightarrow \text{same as CV} = 1$                                                                                                  | No                  | Reformulation                                             |
| 5 | $K = \text{midpoint of } [1/3, 1]$                      | $\text{Midpoint} = (1/3 + 1)/2 = 2/3 \rightarrow \text{same as } K = 2/3$                                                                                         | No                  | Reformulation                                             |
| 6 | Democratic matrix $M = m_0(I + \epsilon D)$             | Eigenvalues of $I + \epsilon D$ are $(1+\epsilon, 1-\epsilon/2, 1-\epsilon/2)$ . Squared masses give specific $K$ . Reduces to $K = 2/3$ at specific $\epsilon$ . | Specific Lagrangian | Reformulation (Lagrangian chosen to reproduce $K = 2/3$ ) |
| 7 | “The masses lie on a circle in $\sqrt{m}$ space”        | Tautological — any three masses do (Section 3)                                                                                                                    | No                  | Tautology                                                 |
| 8 | “The $C_3$ potential ground state selects $120^\circ$ ” | Tautology + saddle (this paper)                                                                                                                                   | No                  | Dead path                                                 |

## J.2: What a Derivation Would Look Like

| Requirement                                   | What It Means                                                                                 | Example (hypothetical)                                                                |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Must produce $a^2 = 2$ specifically           | Not “ $a^2$ is of order 1” or “ $a^2$ is near 2”                                              | A symmetry that forces the Yukawa coupling matrix to have eigenvalue ratio $\sqrt{2}$ |
| Must contain input beyond the three masses    | Something from the Lagrangian, the gauge group, or the vacuum                                 | A property of the Higgs potential or the lepton Yukawa texture                        |
| Must explain or accommodate the quark sectors | Why $a^2 \text{ lep} = 2$ but $a^2 \text{ down} \approx 2.4$ and $a^2 \text{ up} \approx 3.1$ | Different symmetry breaking pattern in quark vs lepton sectors                        |
| Must not be circular                          | Cannot assume $K = 2/3$ and derive $K = 2/3$                                                  | Must start from something that is not equivalent to $K = 2/3$                         |

## APPENDIX K: THE THREE-SECTOR DATA — COMPLETE

### K.1: Masses Used (DATA-3)

| Sector          | $m_1$ (MeV)     | $m_2$ (MeV)       | $m_3$ (MeV)        | Source                                                              |
|-----------------|-----------------|-------------------|--------------------|---------------------------------------------------------------------|
| Charged leptons | $m_e = 0.51100$ | $m_\mu = 105.658$ | $m_\tau = 1776.86$ | CODATA + PDG                                                        |
| Down quarks     | $m_d = 4.7$     | $m_s = 93.5$      | $m_b = 4180$       | PDG (MS-bar at 2 GeV for light quarks, MS-bar at $m_b$ for b)       |
| Up quarks       | $m_u = 2.16$    | $m_c = 1273$      | $m_t = 172570$     | PDG (MS-bar at 2 GeV for u, MS-bar at $m_c$ for c, pole mass for t) |

### K.2: Koide Parameters

| Quantity                            | Charged Leptons | Down Quarks | Up Quarks |
|-------------------------------------|-----------------|-------------|-----------|
| $\sqrt{m_1}$                        | 0.7149          | 2.168       | 1.470     |
| $\sqrt{m_2}$                        | 10.279          | 9.670       | 35.68     |
| $\sqrt{m_3}$                        | 42.153          | 64.65       | 415.4     |
| $\Sigma\sqrt{m}$                    | 53.147          | 76.49       | 452.6     |
| $M = \Sigma\sqrt{m} / 3$            | 17.716          | 25.50       | 150.9     |
| $\Sigma m$                          | 1882.93         | 4278        | 173845    |
| $(\Sigma\sqrt{m})^2$                | 2824.6          | 5850.8      | 204847    |
| $K = \Sigma m / (\Sigma\sqrt{m})^2$ | 0.66666         | 0.7313      | 0.8488    |
| $a^2 = 2(3K - 1)$                   | 2.000           | 2.388       | 3.093     |
| $a$                                 | 1.414           | 1.545       | 1.759     |

### K.3: The Quark Mass Scale Problem

| Issue                             | Description                                                                                                                    |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Quark masses are scheme-dependent | Light quark masses (u, d, s) are MS-bar running masses evaluated at a reference scale. Different scales give different masses. |

| Issue                                                 | Description                                                                                                                                                                             |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The Koide ratio for quarks depends on the scale       | $K(d,s,b)$ changes if the masses are evaluated at different renormalization scales                                                                                                      |
| Lepton masses are pole masses                         | No scheme dependence — the Koide ratio for leptons is unambiguous                                                                                                                       |
| Does $K = 2/3$ hold for quarks at some special scale? | Unknown — this is an open question. If there exists a scale where $a^2 \text{ down} = 2$ or $a^2 \text{ up} = 2$ , that scale might have physical significance. No such scale is known. |

**This is a fundamental asymmetry between the lepton and quark sectors.** Lepton masses are physical (pole) masses with no scheme dependence. Quark masses are running parameters whose values depend on the renormalization scale. The Koide formula for leptons is unambiguous. For quarks, the formula's value depends on the scale at which the masses are evaluated. Any future theory that attempts to derive  $a^2 = 2$  for quarks must specify at which scale the relation is supposed to hold.

## APPENDIX L: THE CONDITIONAL PARAMETER REDUCTION — STATUS

### L.1: The Chain

| Step | Statement                                                  | Status                                                | Paper   |
|------|------------------------------------------------------------|-------------------------------------------------------|---------|
| 1    | SM has 19 free parameters                                  | Established (standard count)                          | —       |
| 2    | $\theta_{\text{QCD}} = 0$ derived from energy minimization | <b>Unconditional</b> — $19 \rightarrow 18$            | PHYS-7  |
| 3    | $K = 2/3$ for charged leptons                              | Observed at $0.91\sigma$ from exact                   | PHYS-8  |
| 4    | IF $K = 2/3$ exact: $m_\tau$ predicted from $m_e, m_\mu$   | <b>Conditional</b> — $18 \rightarrow 17$ if $K$ exact | PHYS-8  |
| 5    | $C_3$ path to derive $K = 2/3$                             | <b>CLOSED</b> (this paper)                            | PHYS-23 |
| 6    | Other derivation paths                                     | Unknown — none known to be viable                     | Open    |
| 7    | Belle II $m_\tau$ precision improvement                    | May test $K = 2/3$ at higher precision                | Future  |

## L.2: What Each Outcome Would Mean

| Outcome                                                              | Effect on Parameter Count                                   | Effect on $K = 2/3$ Status                     |
|----------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------|
| Belle II measures $m_\tau$ consistent with $K = 2/3$ at 7+ sf        | Strengthens conditional; still $18 \rightarrow 17$ if exact | Upgraded from $0.91\sigma$ to sub- $\sigma$    |
| Belle II measures $m_\tau$ inconsistent with $K = 2/3$ at $>3\sigma$ | Conditional closed; stays at 18                             | $K = 2/3$ refuted — was approximate, not exact |
| Someone derives $a^2 = 2$ from physics                               | <b>Unconditional</b> — $18 \rightarrow 17$                  | $K = 2/3$ explained                            |
| No progress                                                          | Conditional remains open                                    | Status unchanged                               |

## L.3: Current Precision Budget

| Quantity                    | Precision (sf) | Limits K Test?                                             |
|-----------------------------|----------------|------------------------------------------------------------|
| $m_e$                       | 9              | No                                                         |
| $m_\mu$                     | 8              | No                                                         |
| $m_\tau$                    | 5              | <b>YES — limiting factor</b>                               |
| K observed                  | 5-6 sf         | Limited by $m_\tau$                                        |
| K predicted (= $2/3$ exact) | $\infty$       | —                                                          |
| Deviation                   | $0.91\sigma$   | Consistent with exact, but cannot rule out $\approx$ exact |

**The entire test rests on  $m_\tau$ .** The electron and muon masses are known to 8-9 significant figures. The tau mass is known to 5. The Koide formula predicts  $m_\tau = 1776.97$  MeV from  $m_e$  and  $m_\mu$ . The measured value is  $1776.86 \pm 0.12$  MeV. The deviation is  $0.11$  MeV =  $0.91\sigma$ . Improving  $m_\tau$  to 6-7 significant figures (reducing the uncertainty from  $0.12$  MeV to  $\sim 0.01$  MeV) would either sharpen the agreement or reveal a discrepancy. Belle II's target precision for  $m_\tau$  is approximately  $0.1$  MeV (comparable to current), with potential improvements from threshold scans reaching  $\sim 0.05$  MeV.

## APPENDIX M: THE CAUCHY-SCHWARZ BOUNDS ON K

### M.1: The Allowed Range

The Koide ratio  $K = \Sigma m / (\Sigma \sqrt{m})^2$  satisfies  $1/N \leq K < 1$  for any  $N$  positive masses. This follows from the Cauchy-Schwarz inequality applied to vectors  $(\sqrt{m_1}, \sqrt{m_2}, \dots, \sqrt{m_N})$  and  $(1, 1, \dots, 1)$ :

$$(\Sigma \sqrt{m_k})^2 \leq N \times \Sigma m_k \rightarrow K \geq 1/N$$

Equality holds when all masses are equal:  $K = 1/N$ .

The upper bound  $K < 1$  follows from  $K = 1$  requiring all mass concentrated in one particle ( $m_1 = M_{\text{total}}$ , all others = 0), which violates positivity of all masses.

### M.2: $K = 2/3$ as the Midpoint

| Bound           | Value ( $N = 3$ )           | Condition                             |
|-----------------|-----------------------------|---------------------------------------|
| Lower           | $K = 1/3$                   | All masses equal: $a = 0$             |
| Upper           | $K \rightarrow 1$           | Extreme hierarchy: one mass dominates |
| <b>Midpoint</b> | <b><math>K = 2/3</math></b> | <b>Equally far from both limits</b>   |

$K = 2/3 = (1/3 + 1)/2$  — exactly the midpoint of the allowed interval  $[1/3, 1)$ . This is Reformulation #5 in the catalog. It sounds significant (“the ratio sits at the center of its allowed range!”) but is algebraically equivalent to  $a^2 = 2$ . The midpoint property does not constitute a derivation.

### M.3: The $a^2 \leftrightarrow K$ Map

| $a^2$                | $K$ (for $N = 3$ ) | Position in $[1/3, 1)$            | Physical Sector        |
|----------------------|--------------------|-----------------------------------|------------------------|
| 0.0                  | 0.333              | Lower bound (equal masses)        | —                      |
| 1.0                  | 0.500              | 1/4 of range                      | —                      |
| <b>2.0</b>           | <b>0.667</b>       | <b>Midpoint</b>                   | <b>Charged leptons</b> |
| 2.388                | 0.731              | 60% of range                      | Down quarks            |
| 3.0                  | 0.833              | 75% of range                      | —                      |
| 3.093                | 0.849              | 77% of range                      | Up quarks              |
| $4/3 \times 3 = 4.0$ | $\rightarrow 1$    | Upper bound (hierarchy saturated) | —                      |

## APPENDIX N: WHAT THE $C_3$ POTENTIAL ACTUALLY DOES AND DOES NOT DO

### N.1: The $C_3$ Frustrated Potential

$$V(\varphi_1, \varphi_2, \varphi_3) = -\sum_i \sum_j \cos(\varphi_i - \varphi_j) = -\cos(\varphi_1 - \varphi_2) - \cos(\varphi_2 - \varphi_3) - \cos(\varphi_1 - \varphi_3)$$

### N.2: What $C_3$ Does

| Property                                    | Achieved?                                | But...                                           |
|---------------------------------------------|------------------------------------------|--------------------------------------------------|
| Selects $120^\circ$ spacing as ground state | Yes — minimally frustrated configuration | The spacing is tautological for any three masses |

| Property                            | Achieved?                                 | But...                                                     |
|-------------------------------------|-------------------------------------------|------------------------------------------------------------|
| Produces a potential with a minimum | Yes — $V$ is minimized at equal spacing   | The minimum of $V$ does not correspond to a minimum of $K$ |
| Is well-motivated physically        | Yes — standard frustrated antiferromagnet | The physical relevance to lepton masses is not established |

### N.3: What $C_3$ Does NOT Do

| Property                                  | Achieved? | Why Not                                                                     |
|-------------------------------------------|-----------|-----------------------------------------------------------------------------|
| Select $a^2 = 2$                          | No        | $V$ has no dependence on the amplitude $a$                                  |
| Select $K = 2/3$                          | No        | $K$ is a saddle at the $V$ -minimum configuration                           |
| Distinguish leptons from quarks           | No        | Same potential for any sector; doesn't explain $a^2$ lep $\neq$ $a^2$ quark |
| Provide a derivation of the Koide formula | No        | Solves the tautological part, fails on the non-trivial part                 |

### N.4: What Would Fix $C_3$

A modified potential  $V' = V + f(a)$  that includes amplitude dependence could in principle select  $a^2 = 2$  if  $f(a)$  has a minimum at  $a = \sqrt{2}$ . But such a potential must be motivated by physics (a symmetry, a coupling structure, a vacuum property), not simply constructed to reproduce the answer. Any  $V'$  that is designed to give  $a^2 = 2$  without independent physical motivation is a reformulation in potential form, not a derivation.

## APPENDIX O: LEVEL 1 / LEVEL 2 — CLASSIFICATION FOR THIS PAPER

### O.1: Level 1 Results (Mathematical, Permanent)

| Result                                                         | Why Level 1                                           |
|----------------------------------------------------------------|-------------------------------------------------------|
| 3 parameters fitting 3 data points = tautology                 | Degree-of-freedom counting — pure mathematics         |
| $\cos(\theta) + \cos(\theta+2\pi/3) + \cos(\theta+4\pi/3) = 0$ | Trigonometric identity — proved from geometric series |
| $K = (1+a^2/2)/3$ for $N = 3$                                  | Trigonometric identity from $\cos^2$ sum = $N/2$      |

| Result                                                              | Why Level 1                                    |
|---------------------------------------------------------------------|------------------------------------------------|
| $K = 2/3 \leftrightarrow a^2 = 2$                                   | Algebraic equivalence                          |
| All 7 reformulations are equivalent                                 | Algebraic identity chain                       |
| $K = 2/3$ is a saddle point of $K(\varphi_1, \varphi_2, \varphi_3)$ | Mathematical property of the function $K$      |
| $C_3$ ground state is $120^\circ$ spacing                           | Standard result for frustrated antiferromagnet |
| Cauchy-Schwarz bounds: $1/3 \leq K < 1$                             | Mathematical inequality                        |
| $K = 2/3$ is the midpoint of $[1/3, 1)$                             | Arithmetic                                     |

## O.2: Level 2 Results (Measured, Could Have Been Different)

| Result                                                                | Why Level 2                            |
|-----------------------------------------------------------------------|----------------------------------------|
| $K = 0.66666$ for charged leptons                                     | Measured from $m_e, m_\mu, m_\tau$     |
| $a^2 = 2.000$ for charged leptons                                     | From $K$ via the Level 1 identity      |
| $a^2 = 2.388$ for down quarks                                         | Measured from $m_d, m_s, m_b$          |
| $a^2 = 3.093$ for up quarks                                           | Measured from $m_u, m_c, m_t$          |
| The ordering $a^2_{\text{lep}} < a^2_{\text{down}} < a^2_{\text{up}}$ | Observed pattern                       |
| $m_\tau = 1776.86 \pm 0.12 \text{ MeV}$                               | Measured (PDG)                         |
| Whether $K = 2/3$ is exact                                            | Requires improved $m_\tau$ measurement |

## O.3: The Closure Is Level 1

The  $C_3$  path is closed by two Level 1 arguments (tautology and saddle point). No measurement is needed to close it. The closure holds in any universe with three positive masses and the Koide parametrization. The amplitude  $a^2 = 2$  being the open problem is a Level 2 question — it depends on our universe having that specific value.

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*Supporting appendix tables H through O for PHYS-23. The tautology is made concrete with explicit construction and five worked examples (Appendix H). The saddle point is demonstrated with numerical values at multiple perturbation sizes (Appendix I). The reformulation catalog is extended with a formal derivation test (Appendix J). The three-sector data is complete with the quark mass scale problem documented (Appendix K). The conditional parameter reduction status is tracked with all possible outcomes (Appendix L). The Cauchy-Schwarz bounds place  $K = 2/3$  as the midpoint of the allowed range — a reformulation, not a derivation (Appendix M). The  $C_3$  potential's exact capabilities and limitations are cataloged (Appendix N). Everything traces to DATA-3 (32/32 pass). The closure is Level 1. The open problem is Level 2.*

[[HOWL-PHYS-23-2026](#)] The Koide  $C_3$  Closure — Tautology, Saddle Point, and the Real Problem. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-23-2026>

[[HOWL-PHYS-1-2026](#)] The Inertial Vortex. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-1-2026>

[[HOWL-PHYS-2-2026](#)] There Are No Constants. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-2-2026>

[[HOWL-PHYS-6-2026](#)] The Confinement Boundary in Integer Arithmetic. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-6-2026>

[[HOWL-PHYS-7-2026](#)] The Strong CP Problem Is Not a Problem. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-7-2026>

[[HOWL-PHYS-8-2026](#)] The Koide Constant Decomposes. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-8-2026>

[[HOWL-PHYS-9-2026](#)] The Electromagnetic Chain in Integer Arithmetic. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-9-2026>

[[HOWL-PHYS-10-2026](#)] Remainder as Observable. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-10-2026>

[[HOWL-PHYS-11-2026](#)] Remainder Structure Across Nine Physics Domains. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-11-2026>

[[HOWL-PHYS-12-2026](#)] Electroweak Integer Anatomy. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-12-2026>

[[HOWL-PHYS-13-2026](#)] Gauge Coupling Unification and Minimal BSM Content. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-13-2026>

[[HOWL-PHYS-14-2026](#)] The Unified Transformation Map. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-14-2026>

[[HOWL-PHYS-15-2026](#)] Integer-Forced Identification of the Minimal Unification Extension. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-15-2026>

[[HOWL-PHYS-17-2026](#)] The Generation Democracy and the Boson Problem. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-17-2026>

[[HOWL-PHYS-18-2026](#)] The  $Y = 1/6$  Asymmetry. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-18-2026>

[[HOWL-PHYS-19-2026](#)] Independent Anomaly Evidence for the Cabibbo Doublet. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-19-2026>



19-2026

[[HOWL-PHYS-20-2026](#)] The Proton Decay Test — Hyper-Kamiokande and the Cabibbo Doublet at  $M_{\text{GUT}} = 10^{15.5}$ . Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-20-2026>

[[HOWL-PHYS-21-2026](#)] The Level 1 / Level 2 Boundary — From Observed Structure to Unobserved Particles. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-21-2026>

[[HOWL-PHYS-22-2026](#)] The  $A_2$  Geometric Cancellation — Rational, Number-Theoretic, and Geometric Anatomy of the QED Two-Loop Coefficient. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-22-2026>